

SCHOOL OF ENGINEERING
COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
B. TECH DEGREE 1ST SEMESTER SECOND INTERNAL EXAMINATIONS, DECEMBER 2023
23-200-0103B: INTRODUCTION TO ELECTRONICS DEVICES & CIRCUITS
(2023 Scheme)- CS,EC,EEE,IT

TIME: 2 hrs

MAX. MARKS: 30

Course Outcomes

On completion of this course the student will be able to:

CO1	To understand the working principle of various semiconductor devices.
CO2	To apply the acquired knowledge to the use of semiconductor devices in various applications.
CO3	To design simple electronic circuits for a given application.

PART A

(Answer all questions)

(5 x 2= 10 Marks)

Q. No.	Questions	Marks	BL	CO	PO
I (a)	What is phase inversion in Common Emitter amplifier? Explain.	2	2	2	1-3,12
(b)	Describe the significance of bypass and coupling capacitors in a Common Emitter amplifier.	2	2	2	1-3,12
(c)	Illustrate and explain the total frequency response of an amplifier and state the equation for its bandwidth.	2	3	2	1-3,12
(d)	Why is Common Collector amplifier called as an Emitter follower?	2	2	2	1-3,12
(e)	A three-stage amplifier has first, second and third stage voltage gains of 10, 15 and 20 respectively. Find the overall voltage gain and its value in decibel.	2	3	2	1-3,12

PART B

(Answer the following questions choosing either the first OR second option)

(1 x 10= 10 Marks)

Q. No.	Questions	Marks	BL	CO	PO
II (a)	Analyse the Common Emitter amplifier configuration with necessary circuit and expressions.	7	4	1	1-3,12
(b)	Explain voltage amplifier using its equivalent circuit.	3	1	1	1-3,12
OR					
III (a)	What are multistage amplifiers and state their purpose? Describe the working of capacitively coupled multistage amplifier with circuit diagram and necessary waveforms.	7	3	1	1-3,12
(b)	An amplifier has a bandwidth of 500 KHz. What is its upper cut-off frequency if the lower cut-off frequency is 25 Hz? Also find the voltage gain at the lower cut-off frequency if the maximum gain is 120.	3	3	2	1-3,12

(1 x 10= 10 Marks)					
IV (a)	Analyze the Common Base amplifier configuration with necessary circuit and expressions.	5	4	1	1-3,12
(b)	Explain the working of FET amplifier with diagram.	5	1	1	1-3,12
OR					
V (a)	Describe in detail the classification of amplifiers based on input and output.	5	4	1	1-3,12
(b)	Analyze the Common Collector amplifier configuration with necessary circuit and expressions.	5	4	1	1-3,12

Bloom's Taxonomy Levels

L1 - %	L2 - %	L3 - %	L4 - %	L5 - %	L6 - %
16%	12%	28%	44%	0%	0%

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